PROBABILITY 3: ASSIGNMENT 7

Problem 1. For each n, let $X_{n,1}, \dots, X_{n,n}$ be independent, with

$$\mathbb{P}(X_{n,k} = \sqrt{n}) = \mathbb{P}(X_{n,k} = -\sqrt{n}) = \frac{1}{2n^2}, \qquad \mathbb{P}(X_{n,k} = 0) = 1 - \frac{1}{n^2}.$$

Put $S_n = \sum_{k=1}^n X_{n,k}$.

(1) Show that the array is centered, $\mathrm{Var}(S_n) = 1$, and the Feller condition holds.

(2) Show that $\mathbb{P}(S_n \neq 0) \leq 1/n$ and deduce $S_n \xrightarrow{\mathbb{P}} 0$. Verify the limit independently using characteristic functions.

Problem 2. Let $0 < a_n \leq n$, with $a_n/n \to 0$. For each n, let $B_{n,1}, \dots, B_{n,n}$ be independent, with

$$B_{n,k} \sim \mathrm{Bernoulli}\left(\frac{a_n k}{n^2}\right), \qquad S_n = \sum_{k=1}^n B_{n,k}, \qquad \mu_n = \mathbb{E}S_n.$$

(1) Compute μ_n and show that $\mathrm{Var}(S_n)/\mu_n \to 1$.

(2) If $a_n \to a \in (0, \infty)$, find the limit law of S_n.

(3) If $a_n \to \infty$, prove $(S_n - \mu_n)/\sqrt{\mu_n} \Rightarrow \mathcal{N}(0,1)$.

Problem 3. Let $Z_1, Z_2, \dots$ be independent, with $|Z_k - \mathbb{E}Z_k| \leq C$ almost surely for a fixed C. Suppose $v_n^2 := \sum_{k=1}^n \mathrm{Var}(Z_k) \to \infty$. Prove that

$$\frac{\sum_{k=1}^n (Z_k - \mathbb{E}Z_k)}{v_n} \Rightarrow N(0,1).$$

Problem 4. For each n, let $B_{n,1}, \dots, B_{n,n}$ be independent, with $B_{n,k} \sim \mathrm{Bernoulli}(p_{n,k})$. Suppose

$$\max_{1 \leq k \leq n} p_{n,k} \longrightarrow 0, \qquad \sum_{k=1}^n p_{n,k} \longrightarrow \lambda \in (0, \infty).$$

Put $S_n = \sum_{k=1}^n B_{n,k}$.

(1) Express the characteristic function $\phi_{S_n}(t)$ as a product. Show that $\sum_k p_{n,k}^2 \to 0$.

(2) Prove $S_n \Rightarrow \mathrm{Poisson}(\lambda)$.

(3) Take $p_{n,k} = k/n^2$. Find the limits in distribution of S_n and $S_n - \mathbb{E}S_n$.